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Statistical Analysis of Damage to Local Businesses Due to Flooding Events Along the Mohawk River Valley in Amsterdam, New York

Tom R. Pascucci, Daniella K. Chernoff, Scott R. Lakeram, and Antonios E. Marsellos

Dept. of Geology, Environment and Sustainability, Hofstra University, Hempstead, NY

Introduction

Amsterdam, New York is located along the banks of the Mohawk River in Upstate New York, between Utica and Albany. The city encompasses numerous business and residential areas which are prone to flash floods during heavy rain events. In late August 2011 Hurricane Irene caused a flash flood giving rise to catastrophic damage to the local area, destroying a museum, numerous business and homes (Leyden 2011). Considering the relative frequency of flooding along the waterway and relative increase in waterflow in the recent past (Cockburn & Garver, 2014), it is necessary to gauge interest that townships have in their surrounding river systems of advanced prognosis for future high water events. In the past, LiDAR data taken from the Mohawk River has been used in conjunction with GIS software to reconstruct past flood events through 3D-simulation (Sisti et.al., 2016). These results both facilitate the reconstruction of previous flooding and eliminate the obligation for first hand site analysis (Sisti et.al., 2016). However, in order to circumvent future damage and possible loss of life, it is necessary to not only reproduce, but predict subsequent areas of high risk prone to flood events and alert residents. A statistical analysis of surveys collected from businesses located near the Mohawk River in Amsterdam, NY is the ultimate goal of this study while simultaneously assessing the preparedness and interest of the local community for online warning systems.

Methodology

A questionnaire was created and refined composed of ten questions, each designed to evaluate different aspects pertaining to the preparation, response and recovery from a flood event in the targeted area. The questions themselves were constructed with a breadth of possible responses to allow the most flexibility of answers while simultaneously providing a firm guideline on a scale of 1 to 5. In the majority of questions a response of 1 signified a “Not Interested” or “Strongly Disagree” and a response of 5 signified a “Very Interested” or “Strongly Agree”.

Once the questionnaire was vetted for optimal relevance and clarity, an area along the Mohawk River was chosen as the proposed study area. Amsterdam was chosen as an average representation of development on both sides of the waterway. Using publically listed addresses a call list was created encompassing businesses within the greater Amsterdam city area along the Mohawk River. The numbers were called using a university phone and every responder was presented with the same scenario describing the project, university affiliation, desire for flood data and security of anonymity. Twenty responses were collated into a data sheet and further statistically analyzed through the use of a Chi-square test and Phi & Cramer’s V measures of association.

Discussion

Of those businesses that have experienced at least one flood event, 40% received the bare minimum of warning prior to a rise in water level, if at all. Additionally, 15% of those businesses that experienced flooding incurred damages of \$5,000.00 or more from the event (Fig. 4, Q4). The vast majority of businesses received little to no damage from flooding (Fig 2, Q1), however there was a relatively large standard deviation (1.35) from the mean response value (1.85). The difference from the mean suggests that the variability of flood damage may be due to business location along the river rather than flood frequency. For those businesses that responded positively to experiencing a flood occurrence at their place of work, it was slightly significant that 38% of them worried of the possibility of an annual flood event (Chi-square test shows p-value=0.076). It appears logical, therefore, that most of the responders encouraged the implementation and utilization of a mobile application concerning the analysis of flood data (Fig.1, Q6-Q8).

A small percentage (25%) of businesses thought that a mobile application that provides some degree of prognosis for the extent of flood water levels would be helpful to their business, but the majority agreed with the necessity of an evacuation route to be provided in advanced or during a flood event (Fig. 4, Q7). This is most likely due to the public dependence of major roadways for both day to day circumstance and as a means

of escape in the event of a natural disaster. Furthermore, some businesses (45%,) agreed that some form of mobile application would speed the recovery process after incurring flood damage (Fig. 4, Q10).

Conclusion

The development of a mobile application is necessary for businesses in the surrounding area of Amsterdam, NY as long as it provides live and relevant flood data that is essential to the continuing vitality of the region. The ability to reference localized flood data enables the municipality to deal with an emergency, such as a flood hazard, in a more efficient and beneficial manner. The survey has shown that a real time flood application would warn local businesses prior to impending floods, provide safe alternate routes for both evacuation and rescue services and allowing business owners with an opportunity for preparedness.

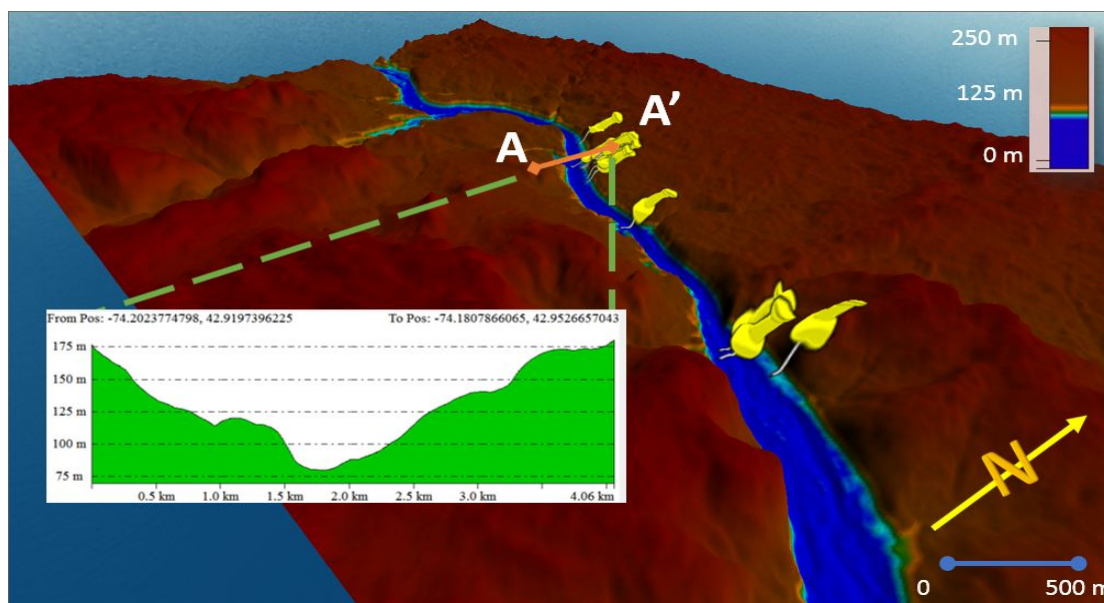


Figure 1. Digital Elevation Model (DEM) of Amsterdam, NY derived from Shuttle Radar Topographic Mission (SRTM) with the locations of the surveyed businesses (yellow pins). The DEM of the study area depicts the location of the businesses surveyed near the Mohawk River. A cross section labeled A to A' depicts the topographic profile of the study in this section of the Mohawk River.

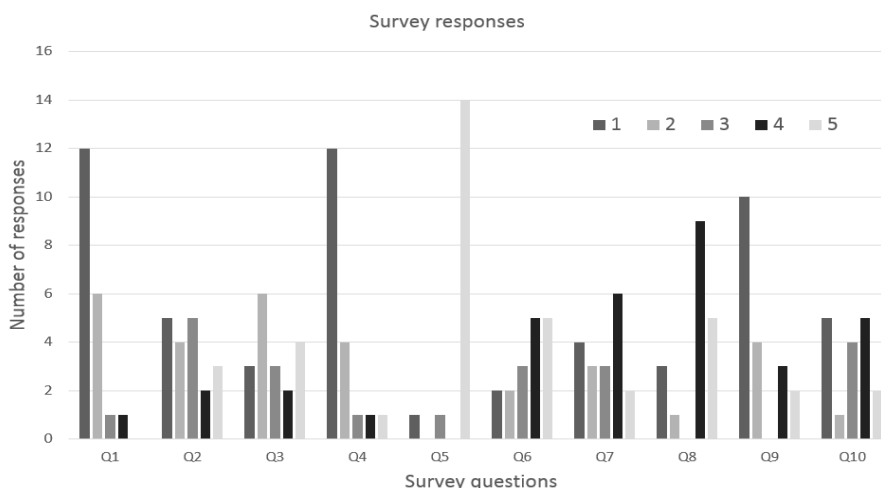


Figure 2. Side-by-side bars chart showing the number of responses from 20 responders for each question. Questions one through five (Q1, Q2, Q3, Q4, Q5) are representative of questions that dealt with flooding events that have occurred in the past and potential flood threats. Questions six through ten (Q6, Q7, Q8, Q9, Q10) discuss the need for a mobile application that could potentially increase flood warning time for business owners and present them with live flood data. Tall bars represent a higher frequency response than lower bars. A full description of the legend can be seen in the text.

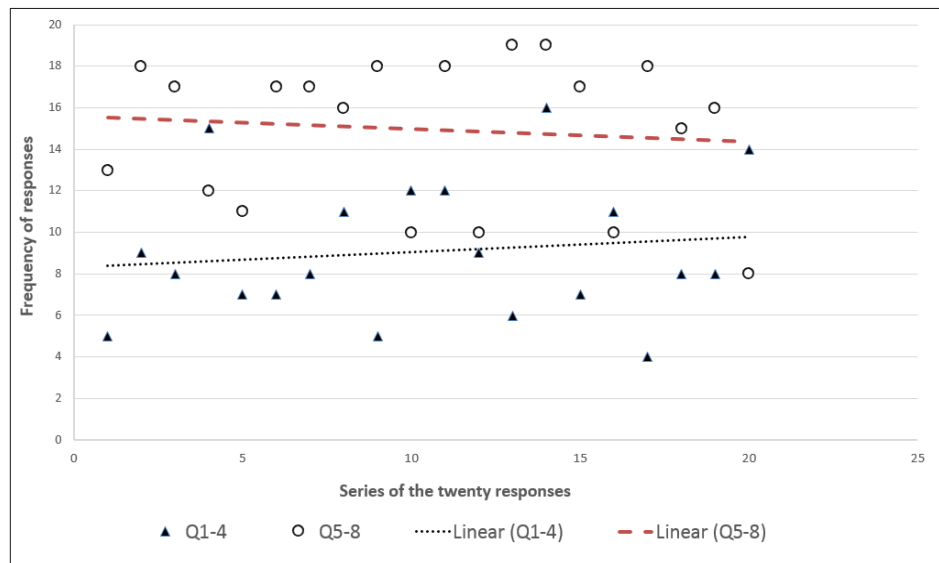


Figure 3. Business Response Distribution. The majority of responses from business in the study area for questions Q1-Q4 display low ranking distribution implying the smaller impact of flooding and damage upon those surveyed. The higher distribution displayed from responses for questions Q5-Q8 implies the need for the application to enhance the resiliency of the business infrastructure and facilitate the speed of recovery following a flood event for a town such as Amsterdam.

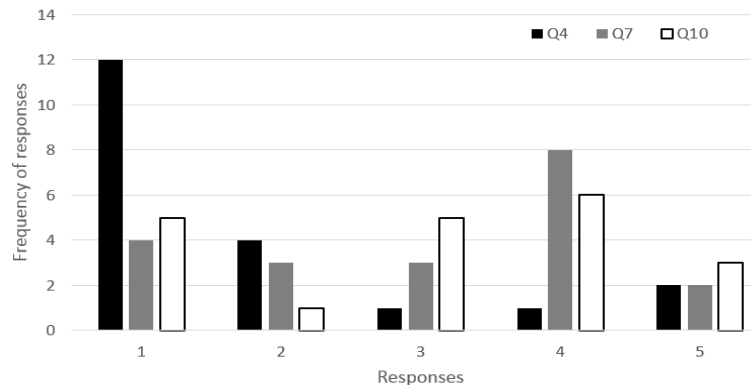


Figure 4. Public Response to Questions of Flood Hazard Impact. Data compiled from responses in a continuous range from 1-5. Question 4 (Q4) examined responses of damages incurred from 1-no damage, to 5- more than \$10,000 in damage. Questions 7 and 10 were scaled 1-5 on responder interest from 1, strongly disagree, to 5 strongly agree; Q7 examined the need for a dynamic evacuation route during a flood hazard event, Q10 explored the impact of an application on flood recovery (N=20, mean=1.85(Q4), 3.05(Q7), 3.05(Q10)).

Cited References

Cockburn, J., Garver, J.I. 2015. Abrupt change in runoff on the north slope of the Catskill Mountains, NY, USA: Above average discharge in the last two decades. *Journal of Hydrology: Regional Studies*: 3, 199-210.
 Leyden, Liz, 01 Sept. 2011. Manor That Has Stood for Centuries Teeters in Storm's Wake." *The New York Times*. The New York Times

Sisti, A., Combs, E., Marsellos, A.E. 2016. Flooding of the Mohawk River at Lock 12 in Fort Hunter, NY, during Hurricane Irene (August 28-29th, 2011). *Proceedings from the Mohawk Watershed Symposium 2016*, p. 45-47, ISBN: 978-1-939968-07-4.

Poster Presentation